Three-copy undistillability of qutrit Werner states by an intrinsic positive identity

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Abstract

We present a finite-certificate proof of three-copy undistillability for the qutrit Werner state $(2I_9 - F)/15$, where F exchanges the two qutrits. Equivalently, an alternating partial-trace quadratic form is nonnegative on every complex 27×27 matrix of rank at most two, with no normality or common-support assumption. The proof is expressed intrinsically through the second compound $M(C) = \bigwedge^2 C$. Three fixed positive Gram operators give a sum-of-squares identity for $\|M(C)\|_F^4 q_{1/2}(C)$. Physical zeros determine the Gram supports; exact range compatibility, a rational positive trial and a certified physical singular-value gap permit an exact positive correction. We isolate these finite inputs and prove a sharper correction bound, leaving an explicit mixed-sector reserve greater than $95/127$ of its certified trial threshold. The endpoint implies the sharp minimum $(1-2\alpha)(1-\alpha)^2$ on $0 \le \alpha \le 1/2$ and the exact three-copy distillability threshold of the physical qutrit Werner family. For every fixed pair of support planes, we give four nonnegative squared moduli and a quantitative description of a possible zero line. The computational premises and their verification status are separated from the analytic implications.

Draft verification status. The finite inputs in Proposition 6 are taken from the existing certificate record [19]; this article does not claim a new independent replay. The complete listed local inventory has been checked for availability and hash agreement. A public immutable release and an external mathematical replay are not claimed. The data availability statement records this boundary.

1 Introduction

Entanglement distillation asks whether noisy bipartite entanglement can be converted, by local operations and classical communication, into an entangled two-qubit state. Positivity of the partial transpose precludes distillation. The converse remains a central structural question: can a state with negative partial transpose remain undistillable at every finite copy number? Werner states provide a canonical symmetric setting for this question [1, 2, 3]. A finite-copy result has an exact operational meaning, while retaining a distinction from all-copy undistillability.

The Schmidt-rank-two criterion converts the physical problem into a matrix inequality. Costa Rico developed the arbitrary-copy partial-trace formulation [4]. Recent two-copy results establish coincidence of the one-copy and two-copy thresholds [5, 6, 7, 8]. The rank-r partial-trace theorem of Fraser, Huber, Pozsgay and Vona [7] supplies the inequality used for our sharp interval continuation. At three copies, Wu and Zou prove endpoint positivity for normal rank-two matrices in arbitrary local dimensions, together with specified nonnormal sectors determined by local support overlap or a product ray in a support plane [9]. The contribution considered here is the full complex rank-two domain when all three local coefficient spaces have dimension three. The two supporting planes are independent, and the matrices need not be normal or Hermitian.

The obstruction to a direct quadratic proof is geometric. The energy form is indefinite on the ambient matrix space, but the allowed rank-two matrices form a nonlinear variety. The second compound matrix records the two-dimensional row and column geometry simultaneously. If $C = XY^*$ with two-column factors, then

$$\bigwedge^2 C = (x_0 \wedge x_1)(y_0 \wedge y_1)^*.$$

This identity permits an exact transport of a positive polynomial Gram construction into features depending only on C and its actual minors [20]. The transport retains complete complex quadratic forms, rather than bounding their cross terms separately.

The positive operators arise from a finite construction. Known physical zeros prescribe exact kernels. Derivatives of these same zeros produce coefficient relations, which establish compatibility of the target with a supported transfer map. A rational trial is then corrected on its prescribed faces. Symmetry reduction and rational sum-of-squares recovery have established precedents [10, 11]; the task-specific content is the physical carrier, its forced kernels, the compatible range and the norm control of the complete correction.

We organize the argument around an explicit distinction. The analytic theorems show exactly how specified finite predicates imply the endpoint and its consequences. Proposition 6 states the computational inputs from the existing proof record, and Appendix E describes their mathematical acceptance predicates. Hash agreement identifies those data; it does not verify their mathematical content.

The paired two-qubit construction comes after global positivity. For any two actual support planes, the compressed swap operator is strictly positive, and the endpoint makes its logical partial transpose positive. Two-qubit separability and product-length theorems [12, 13, 14, 15] then apply. We give bounded filters, explicit product factors once a normal form is chosen, and a spectral argument showing that each pair has at most one zero amplitude up to complex scaling. These local four-term formulas depend on the support planes; they do not replace the universal polynomial certificate.

2 Conventions and main statements

Let $\mathcal{H} = (\mathbb{C}^3)^{\otimes 3}$, with fixed computational product bases. The tensor factors label resource copies, not three communicating parties. The physical bipartition is $A_1A_2A_3 : B_1B_2B_3$. All inner products are conjugate-linear in the first argument, and all partial traces are unnormalized. Define

$$q_\alpha(C) = \sum_{J\subseteq\{1,2,3\}} (-\alpha)^{|J|} \|\mathrm{Tr}_J\, C\|_F^2, \qquad q = q_{1/2}. \tag{2.1}$$

Here $\mathrm{Tr}_\varnothing C = C$. Thus, writing $N = \|C\|_F^2$, $A_1 = \sum_i \|\mathrm{Tr}_i C\|_F^2$, $B_2 = \sum_{i<j} \|\mathrm{Tr}_{ij} C\|_F^2$ and $T = |\operatorname{Tr} C|^2$,

$$q(C) = N - \frac{1}{2}A_1 + \frac{1}{4}B_2 - \frac{1}{8}T. \tag{2.2}$$

Theorem 1 (Three-copy qutrit endpoint)**.** *For every $C \in \mathrm{End}(\mathcal{H})$ with $\operatorname{rank} C \le 2$,*

$$q(C) \ge 0. \tag{2.3}$$

The proof uses the finite certificate inputs of Proposition 6, in their original physical metrics.

Corollary 2 (Sharp interval and physical threshold)**.** *For $0 \le \alpha \le 1/2$,*

$$\min_{\operatorname{rank} C\le 2,\ \|C\|_F=1} q_\alpha(C) = (1-2\alpha)(1-\alpha)^2. \tag{2.4}$$

An attainer is $C_\star = \mathrm{diag}(1,1,0)/\sqrt{2} \otimes |0\rangle\langle 0| \otimes |0\rangle\langle 0|$. *For the physical qutrit Werner family*

$$\rho_\alpha = \frac{I_9 - \alpha F}{9 - 3\alpha}, \qquad -1 \le \alpha \le 1, \tag{2.5}$$

three-copy distillability holds exactly when $\alpha > 1/2$. *The NPT yet three-copy-undistillable interval is* $1/3 < \alpha \le 1/2$.

To verify the physical equivalence, use row-first vectorization,

$$\mathrm{vec}\, C = \sum_{a,b} C_{ab} |a\rangle_A |b\rangle_B.$$

An SVD gives $\mathrm{vec}\, C = \sum_j s_j u_j \otimes \overline{v_j}$, so the Schmidt rank equals $\mathrm{rank}\, C$. With $|I_3\rangle = \sum_{j=0}^{2} |jj\rangle$ and $\Omega = |I_3\rangle\langle I_3|$, one has $F^\Gamma = \Omega$. Contraction with $\langle I_3|$ at site i gives $\mathrm{Tr}_i\, C$, hence

$$\langle \mathrm{vec}\, C, (\rho_\alpha^\Gamma)^{\otimes 3}\, \mathrm{vec}\, C\rangle = \frac{q_\alpha(C)}{(9-3\alpha)^3}. \tag{2.6}$$

The denominator is positive throughout the physical interval. The eigenvalue of ρ_α^Γ on $|I_3\rangle/\sqrt{3}$ is $(1-3\alpha)/(9-3\alpha)$, equal to $-1/15$ at the half endpoint. These conventions differ in the sign of α from the $I + \alpha F$ convention used in several cited works.

The rank restriction is essential: $I_3/\sqrt{3} \otimes |0\rangle\langle 0| \otimes |0\rangle\langle 0|$ has rank three, norm one and energy $-1/8$.

3 An intrinsic compound identity

Equip $\bigwedge^2 \mathcal{H}$ with the orthonormal basis $e_i \wedge e_j = (e_i \otimes e_j - e_j \otimes e_i)/\sqrt{2}$, $i < j$. For

$$M(C) = \bigwedge^2 C, \qquad M_{(ij),(kl)} = C_{ik}C_{jl} - C_{il}C_{jk}, \qquad \Delta(C) = \|M(C)\|_F^2, \tag{3.1}$$

the singular values of $M(C)$ are $s_i s_j$, $i < j$. Therefore

$$\Delta(C) = \frac{(\mathrm{Tr}\, C^*C)^2 - \mathrm{Tr}((C^*C)^2)}{2}. \tag{3.2}$$

At rank two, $\Delta = s_1^2 s_2^2 > 0$; at rank at most one, $\Delta = 0$.

Let $\mathcal{E} = \mathcal{H} \otimes \bigwedge^2 \mathcal{H} \otimes \bigwedge^2 \mathcal{H} \otimes \mathcal{H}$ and define $Z(C; a, b)_{u,I,K,v} = C_{uv} a_I b_K$. The second and third factors are exterior coordinates; the matrix column is in the final slot.

Theorem 3 (Intrinsic positive identity)**.** *The finite certificate supplies fixed* $G_h, G_m, G_a \succeq 0$ *on* $\mathcal{E}$, *extended by zero from the carriers defined below. For every* $\mathrm{rank}\, C \le 2$, *define*

$$f_{h,JK} = Z(C; M_J, M_K), \qquad f_{m,JK} = Z(C; M_J, \overline{M_K}), \qquad f_{a,JK} = Z(C; \overline{M_J}, \overline{M_K}). \tag{3.3}$$

Here J, K *run over the 351 exterior coordinates. Then*

$$\Delta(C)^2 q(C) = \sum_{s \in \{h,m,a\}} \sum_{J,K} f_{s,JK}^* G_s f_{s,JK}. \tag{3.4}$$

Factoring $G_s = R_s^* R_s$ *expresses the right side as squared moduli of polynomials of total degree five.*

Proof. At rank two write $C = XY^*$ with two full-column-rank factors. Set $p = x_0 \wedge x_1$, $r = y_0 \wedge y_1$ and $g_X = \|p\|^2$, $g_Y = \|r\|^2$. Cauchy–Binet gives

$$M = pr^*, \qquad M_J = p\,\overline{r_J}, \qquad \Delta = g_X g_Y. \tag{3.5}$$

Both shore incidences hold: $p \wedge C_{\cdot j} = 0$ and $\overline{r} \wedge C_{i\cdot} = 0$, with the row regarded as a coordinate vector.

The Gram construction of Sections 4–5 gives

$$\begin{aligned} z_h &= Z(C; p, p), \qquad z_m = Z(C; p, \overline{p}), \qquad z_a = Z(C; \overline{p}, \overline{p}), \\ g_X^2 q(C) &= \sum_s z_s^* G_s z_s. \end{aligned} \tag{3.6}$$

The complete feature identities are

$$f_{h,JK} = \overline{r_J}\,\overline{r_K} z_h, \quad f_{m,JK} = \overline{r_J} r_K z_m, \quad f_{a,JK} = r_J r_K z_a.$$

Each quadratic form, including all its internal cross terms, is multiplied by $|r_J|^2|r_K|^2$. Consequently

$$\sum_{s,J,K} f_{s,JK}^* G_s f_{s,JK} = \left(\sum_J |r_J|^2\right)^2 \sum_s z_s^* G_s z_s = g_Y^2 g_X^2 q(C) = \Delta^2 q(C).$$

At rank at most one both sides vanish. The bidegrees of the three features in $(C, \overline{C})$ are $(5,0), (3,2), (1,4)$. □

This transport is independent of a source frame. Under $X \mapsto XS$, $Y \mapsto YS^{-*}$, $p \mapsto (\det S)p$ and $r \mapsto (\overline{\det S})^{-1} r$, so M and Δ are unchanged. The three source features acquire factors $(\det S)^2$, $|\det S|^2$ and $\overline{\det S}^{\,2}$, respectively. Every source evaluation scales by $|\det S|^4$, exactly as $g_X^2 q$.

For rank two, (3.4) implies the endpoint by division by $\Delta^2 > 0$. At rank one, $C = uv^*$ satisfies the direct identity

$$q(uv^*) = \langle u \otimes v, \prod_{i=1}^{3} (I - F_i/2)\, u \otimes v \rangle \geq \frac{1}{8}\|u\|^2\|v\|^2. \tag{3.7}$$

The three swaps commute and have eigenvalues ± 1. This proves the rank-one case without division by a vanishing determinant. An alternative is to extend $\operatorname{ran} C$ to an orthonormal two-plane in (3.6), making $g_X = 1$.

The intrinsic identity inherits the certified Gram operators. It does not constitute an independent proof of their positivity.

4 The finite Gram construction

We use six global $\mathcal{H}$ slots ordered $(C \text{ row}, p_1, p_2, p_1', p_2', C \text{ column}) = (0, 1, 2, 3, 4, 5)$. Put $A_{ij} = (I - F_{ij})/2$, and let Alt_{ijk} be the orthogonal three-slot antisymmetrizer. If E_λ denotes the central Young projector, define

$$\Pi_h = E_{(3,2)}^{(01234)} A_{12} A_{34}, \qquad \Pi_m = (A_{12} - \mathrm{Alt}_{012}) A_{34}, \qquad \Pi_a = E_{(2,2)}^{(1234)} A_{12} A_{34}. \tag{4.1}$$

Lemma 4 (Carrier and transpose identities)**.** *The source features satisfy $\Pi_s z_s = z_s$. Let Γ_m transpose the entire second minor subsystem $3, 4$, and let Γ_a transpose the entire subsystem $1, 2, 3, 4$. For every operator G on the corresponding full tensor space,*

$$z_s^* G z_s = z_h^* \Gamma_s(G) z_h, \qquad s = m, a. \tag{4.2}$$

Proof. The pair antisymmetries are immediate. The decomposable tensor $p \otimes p$ belongs to the Schur module $S_{(2,2)}\mathcal{H}$. For a column $c \in \mathrm{span}\{x_0, x_1\}$, $c \otimes p \otimes p$ lies in $S_{(3,2)}\mathcal{H}$; equivalently, the incidence $p \wedge c = 0$ removes the three-row component. The mixed feature satisfies $\mathrm{Alt}_{012}(c \otimes p) = 0$. Appendix A gives direct highest-weight and finite-projector verifications of these inclusions.

For a fixed split $a \otimes b$, expand

$$(a \otimes \overline{b})^* G (a \otimes \overline{b}) = \sum_{i,j,k,l} \overline{a_i} b_j G_{ij,kl} a_k \overline{b_l}.$$

Interchanging j, l gives the expectation of $\Gamma_b(G)$ on $a \otimes b$. Apply this to the indicated minor factors. The matrix coordinate C_{uv} is unchanged. □

Define

$$T_0 = \prod_{i=1}^{3} (I - \Omega_{0_i 5_i}/2), \quad \overline{T} = \Pi_h T_0 \Pi_h, \quad \mathcal{A}_s^*(G) = \Pi_h \Gamma_s(G) \Pi_h. \tag{4.3}$$

The full target identity is

$$\overline{T} = G_h + \mathcal{A}_m^*(G_m) + \mathcal{A}_a^*(G_a), \qquad G_s \succeq 0. \tag{4.4}$$

Transpose precedes compression. It preserves the Hilbert–Schmidt norm, not positivity. Evaluation on z_h, using $z_h^* T_0 z_h = g_X^2 q(C)$, gives (3.6) with all eight subset terms and every complex phase retained.

4.1 Symmetry and the physical metric

Local-unitary conjugation of C, simultaneous site permutations and complex conjugation preserve the target. Averaging a feasible Gram tuple over these symmetries preserves positivity and the full identity. The construction uses a real invariant block ansatz; it need only exhibit one sufficient certificate. The local actions in sectors h, m, a have unbarred slot sets 01234, 012, 0, respectively, with conjugate action on the remaining slots. These actions are intertwined by the whole-minor transpose maps.

In a physical isotypic decomposition, a block G_b of multiplicity ν_b contributes $\nu_b \operatorname{Tr}(G_b^2)$ to the HS norm. If V_b is a nonorthonormal native frame, its Gram matrix $H_b = V_b^* V_b$ must be retained. For $G_b = V_b Z V_b^*$,

$$\operatorname{Tr}(G_b^2) = \operatorname{Tr}(Z H_b Z H_b).$$

No Euclidean coefficient norm replaces this physical quantity.

4.2 Zeros, faces and exact range

For any directly verified zero C_0, positivity of (3.6) forces $G_s z_s(C_0) = 0$ in every sector. A covariance of zero features therefore defines a necessary support $P_s \le \Pi_s$ for every feasible G_s:

$$G_s = P_s G_s P_s.$$

The construction intersects the kernels from product-spectator, binary spin and qutrit-spectator zero families. Their exact definitions and nine selected factorizations are in Appendix B. The original 414 symmetry blocks leave 327 nonempty faces. Restriction to these faces is necessary; it does not by itself prove feasibility.

At a selected zero, vary both factors of $C = XB$ through arbitrary complex directions. With $J_h = \dot{z}_h z_h^* + z_h \dot{z}_h^*$ and $\Gamma_h = \mathrm{Id}$,

$$P_s \Gamma_s(J_h) P_s = 0. \tag{4.5}$$

Both real and imaginary variations are included. Direct full-qutrit checks of $X^*\mathcal{L}(C_0) = 0$ and $\mathcal{L}(C_0)B^* = 0$ give $\mathrm{Tr}(J_h\overline{T}) = \mathrm{d}(g_X^2 q)(C_0) = 0$, where $\mathcal{L} = (D \mapsto D - \mathrm{Tr}(D)I_3/2)^{\otimes 3}$. Thus the relations annihilate the target without assuming universal endpoint positivity.

After eliminating the supported h coefficients, let A be the rational transfer with 2463 rows and t its target. The finite range predicates are

$$L\mathsf{A} = 0, \quad Lt = 0, \quad \operatorname{rank} L = 181, \quad \operatorname{rank} \mathsf{A} \geq 2282. \tag{4.6}$$

They imply $\operatorname{ran} \mathsf{A} = \ker L$ and $\operatorname{rank} \mathsf{A} = 2282$. Nonzero designated minors modulo 65521 certify independence after checking denominator invertibility. The rational equations, not modular rank alone, prove target compatibility. The correction domain retains the designated independent columns.

5 Positive repair and the finite certificate

The quantitative refinements below follow the publication review [21]; their derivations are included. All operator spaces in this section are real Hilbert spaces of self-adjoint operators with their physical HS inner products. Let $\mathcal{F}_h(H) = P_h H P_h$, $\mathcal{Q}_h = I - \mathcal{F}_h$ and

$$\mathcal{C}(u_m, u_a) = \mathcal{A}_m^*(u_m) + \mathcal{A}_a^*(u_a), \qquad \mathcal{D} = \mathcal{Q}_h \mathcal{C}|_{\mathcal{U}}.$$

The correction domain $\mathcal{U}$ is supported on the m, a faces.

Lemma 5 (Orthogonal positive repair)**.** *Let F be an orthogonal projection, $Q = I - F$, $D = Q\mathcal{C}$ and $\|\mathcal{C}\| \leq \kappa$. Suppose $QE \in \operatorname{ran} D$, $\|E\| < \varepsilon$ and $0 < \sigma < \sigma_{\min}^+(D)$. Then*

$$u = D^\dagger QE, \qquad v = E - \mathcal{C}u \quad \textit{satisfy} \quad Qv = 0, \quad \|u\| < \frac{\varepsilon}{\sigma}, \quad \|v\| < \frac{\kappa\varepsilon}{\sigma}. \tag{5.1}$$

Proof. Compatibility gives $Du = QE$. Set $x = \|QE\|$, $y = \|FE\|$. The supported inverse gives $\|u\| \leq x/\sigma$. Orthogonality yields

$$\|F\mathcal{C}u\|^2 = \|\mathcal{C}u\|^2 - \|Q\mathcal{C}u\|^2 \leq (\kappa^2/\sigma^2 - 1)x^2.$$

Since $v = F(E - \mathcal{C}u)$, Cauchy–Schwarz gives

$$\|v\| \leq y + \sqrt{\kappa^2/\sigma^2 - 1}\, x \leq \frac{\kappa}{\sigma}\sqrt{x^2 + y^2} < \frac{\kappa\varepsilon}{\sigma}.$$

The bracket is nonnegative because $\sigma < \sigma_{\min}^+(D) \leq \|D\| \leq \kappa$. □

Let the supported positive trial be G_s^0 and retain its complete defect

$$E = \overline{T} - G_h^0 - \mathcal{A}_m^*(G_m^0) - \mathcal{A}_a^*(G_a^0). \tag{5.2}$$

Proposition 6 (Finite certificate inputs)**.** *The existing certificate record [19] supplies the following finite assertions for the operators and physical coordinates just defined:*

(i) The face projectors and coordinate dictionaries give the stated 414 blocks, 327 nonempty faces and complete full-transpose maps.

(ii) The exact range predicates (4.6) hold. The 11149-dimensional domain $\mathcal{U}$ contains the original 2282 independent columns, and $\mathcal{Q}_h E \in \operatorname{ran} \mathcal{D}$.

(iii) Every nonempty trial face has weighted margin greater than 2^{-13}, 2^{-18}, 2^{-15} in sectors h, m, a, respectively, and $\|E\|_{\mathrm{HS}} < 2^{-35}$ on the complete 122-block target.

(iv) A specified dyadic matrix A_{48} satisfies $\|\mathcal{D} - A_{48}\|_{\mathrm{op}} < 2^{-32}$ in physical orthonormal coordinates. All 664 transfer blocks are included.

(v) The symmetric entrywise grid-rounding S_{48} of $A_{48}A_{48}^*$ satisfies $\|A_{48}A_{48}^* - S_{48}\|_{\mathrm{op}} < 2463\,2^{-48}$ and $\lambda_{182}(S_{48}) > 2^{-30}$, with eigenvalues increasing.

These are attributed computational inputs, not conclusions of the small diagnostic examples below. Their acceptance predicates and present verification status are specified in Appendix E.

Lemma 7 (Sharpened physical gap)**.** *The finite inputs imply*

$$\sigma_{\min}^{+}(\mathcal{D}) > \sqrt{259681}\,2^{-24} - 2^{-32} > 127\,2^{-22}. \tag{5.3}$$

Proof. Eigenvalue perturbation gives $s_{182}(A_{48})^2 > 2^{-30} - 2463\,2^{-48} = 259681\,2^{-48}$. Singular-value perturbation subtracts at most 2^{-32}. Exact rank 2282 in 2463 output dimensions makes the 182nd singular value the least positive one. The last comparison follows by squaring positive quantities:

$$259681 - (508 + 1/256)^2 = 105711615/65536 > 0.$$

□

Full transpose is an HS isometry and compression a contraction, so $\|\mathcal{C}\| \le \sqrt{2}$. Apply Lemma 5 with $\varepsilon = 2^{-35}$ and $\sigma = 127\,2^{-22}$:

$$\begin{aligned} u &= \mathcal{D}^{\dagger}\mathcal{Q}_h E, \\ G_m = G_m^0 + u_m, \quad G_a &= G_a^0 + u_a, \quad G_h = G_h^0 + E - \mathcal{C}u. \end{aligned} \tag{5.4}$$

The supports are preserved because $\mathcal{Q}_h(E - \mathcal{C}u) = 0$, and (4.4) holds by substitution. With $b = 2^{-18}$, the corrections obey $\|u\| < (32/127)b$ and $\|G_h - G_h^0\| < (32\sqrt{2}/127)b < (48/127)b$. Since $\sqrt{\nu_b}\|\delta G_b\|_{\mathrm{op}} \le \|\delta G_s\|_{\mathrm{HS}}$, every face remains positive with the explicit margins in Table 1.

Sector	Trial weighted margin	Correction bound	Corrected margin
h	$> 32b$	$< (48/127)b$	$> (4016/127)b$
m	$> b$	$< (32/127)b$	$> (95/127)b$
a	$> 8b$	$< (32/127)b$	$> (984/127)b$

Table 1: All nonempty faces, with $b = 2^{-18}$ and original physical multiplicities. Empty faces remain empty. The improved bounds use the existing finite inputs without another large computation.

The exact correction has a rational native specification. If E_j are native source atoms, $M_{\mathcal{U},ij} = \mathrm{Tr}(E_iE_j)$, and D_I, e_I restrict the transfer and defect to the independent rows, then

$$c = M_{\mathcal{U}}^{-1}D_I^T(D_IM_{\mathcal{U}}^{-1}D_I^T)^{-1}e_I, \qquad u = \sum_j c_jE_j. \tag{5.5}$$

Both inverses are ordinary positive inverses. Lagrange multipliers identify this as the unique minimum of $c^TM_{\mathcal{U}}c$ subject to $D_Ic = e_I$. Exact compatibility supplies every remaining row. An invertible change of output coordinates leaves this constrained minimum unchanged; the source metric, including its cross entries, cannot be omitted.

Equations (5.4) and (5.5) therefore specify an exact positive tuple. Lemma 4 gives (3.6); Theorem 3 and (3.7) complete Theorem 1.

6 Sharp continuation and the Werner threshold

For the same rank-two matrix throughout, set

$$R_3 = N - \frac{T}{2}, \qquad R_{12} = N - \frac{A_1 + B_2}{6} + \frac{T}{4}. \tag{6.1}$$

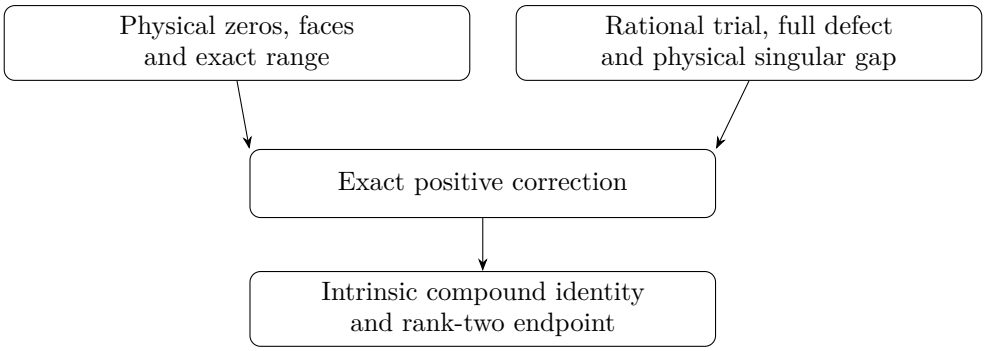


Figure 1: The range and quantitative branches meet on the same physical operators. The paired four-square representation is a consequence of the endpoint, not an input to this diagram.

The nuclear trace bound gives $R_3 \geq 0$. The rank-two instance of the partial-trace inequality of Fraser et al. [7] is

$$\|\operatorname{Tr}_L C\|_F^2 + \|\operatorname{Tr}_R C\|_F^2 \leq 2\|C\|_F^2 + \frac{1}{2}|\operatorname{Tr} C|^2. \tag{6.2}$$

Apply it on each of the three actual 3-by-9 regroupings of C and sum, obtaining $R_{12} \geq 0$. Regrouping preserves global rank; a partial trace need not preserve it. A direct proof of the needed rank-two inequality is included in Appendix C.

Coefficient comparison gives the exact identity

$$\begin{aligned} q_\alpha(C) = (1-2\alpha)(1-\alpha)^2 N + \frac{4\alpha(1+\alpha)}{3} q(C) \\ + \alpha(1-2\alpha)(2/3-\alpha)R_3 + 2\alpha(1-2\alpha)R_{12}. \end{aligned} \tag{6.3}$$

All coefficients are nonnegative on $[0, 1/2]$. The matrix $C_\star$ in Corollary 2 has invariants $(N, A_1, B_2, T) = (1, 4, 5, 2)$ and attains the claimed minimum.

For $\alpha \leq 0$, each factor $I - \alpha\Omega$ is positive. For $\alpha > 1/2$, the rank-two unit matrix

$$C_> = \frac{\operatorname{diag}(1,1,0)}{\sqrt{2}} \otimes |0\rangle\langle 1| \otimes |0\rangle\langle 1| \tag{6.4}$$

has invariants $(1, 2, 0, 0)$ and energy $q_\alpha(C_>) = 1 - 2\alpha < 0$. This remains negative at $\alpha = 1$. Together with (2.6), these observations prove the full physical threshold. Padding a lower-copy witness by $|0\rangle\langle 0|$ multiplies its energy by $1 - \alpha$ and preserves rank, so endpoint nonnegativity also includes the one-copy and two-copy tests.

7 Consequences on fixed support planes

Let $U, V : \mathbb{C}^2 \to \mathcal{H}$ be arbitrary isometries. For $C = UAV^*$, $A \in M_2(\mathbb{C})$, define

$$D = (U \otimes V)^* \prod_{i=1}^{3} (I - F_i/2)(U \otimes V), \qquad K = D^\Gamma. \tag{7.1}$$

The swaps act between the two global replicas, and Γ transposes the second logical qubit. One always has $D \succeq I_4/8$. In logical coordinates, $K_{ab,cd} = D_{ad,cb}$. Contracting with $\overline{A_{ab}} A_{cd}$ gives

$$q(UAV^*) = (\operatorname{vec} A)^* K \operatorname{vec} A. \tag{7.2}$$

Theorem 1, applied to every amplitude A, therefore gives $K \succeq 0$.

7.1 An unconditional spectral gap and zero geometry

Proposition 8 (Fixed-plane geometry)**.** *With eigenvalues increasing,* $\lambda_2(K) \geq 1/8$ *independently of the endpoint theorem. In particular* $\dim\ker K \leq 1$*. Under endpoint positivity, one of the four product frames* $I\otimes I$*,* $I\otimes\sigma_1$*,* $\sigma_1\otimes I$*,* $\sigma_1\otimes\sigma_1$ *has a leading* 3×3 *compression* $H \succeq I_3/32$*. If* K *is singular, its physical zero set*

$$\mathcal{Z}_{U,V} = \{UA_0V^* : \operatorname{vec} A_0 \in \ker K\}$$

is a complex line and

$$q(UAV^*) \geq \frac{1}{8}\operatorname{dist}_F(UAV^*, \mathcal{Z}_{U,V})^2. \tag{7.3}$$

Every nonzero element of that line has rank two.

Proof. Every unit rank-one amplitude gives a rank-one physical matrix and hence energy at least $1/8$ by (3.7). Every two-dimensional complex subspace of M_2 contains a nonzero singular matrix: the homogeneous determinant polynomial on its projective line either vanishes identically or has a root. If $\lambda_2(K) < 1/8$, the two lowest eigenspaces contradict this rank-one floor.

When $K \succeq 0$, choose a unit lowest eigenvector ψ. Then $K \succeq (I - \psi\psi^*)/8$. Some coordinate satisfies $|\psi_j|^2 \geq 1/4$; on the three-coordinate complement of that basis vector the lower bound is at least $I_3/32$. The four bit-flip frames can move this omitted coordinate to the last position. Finally, if K is singular, its three nonzero eigenvalues are at least $1/8$. Orthogonal projection onto its kernel and the isometry $A \mapsto UAV^*$ prove (7.3). A rank-one kernel vector is excluded by the same floor. □

For positive definite K with $\lambda_1(K) < 1/8$, the zero set is $\{0\}$ and the correct distance coefficient is $\lambda_1(K)$. Thus (7.3) is not asserted uniformly over all plane pairs.

Strict positivity on the corrected Gram faces also gives a different quantitative zero statement. In an orthonormal source frame, let $z_{s,b}^{\mathrm{supp}}$ be the supported feature in the complete physical isotypic block of multiplicity ν_b. Then

$$q(C) \geq \frac{95}{127}2^{-18}\sum_{s,b}\nu_b^{-1/2}\|z_{s,b}^{\mathrm{supp}}\|^2. \tag{7.4}$$

The norms include the entire physical block; the multiplicity is not inserted a second time. Vanishing of all supported features is equivalent to $q(C) = 0$. An inverse theorem for the nonlinear feature map would be needed to turn (7.4) into a global distance estimate.

7.2 Four product factors and four squared moduli

Because $D > 0$ and $D^\Gamma \succeq 0$, two-qubit PPT sufficiency makes $D/\operatorname{Tr} D$ separable [12, 13]. The known two-qubit product-length results [14, 15] yield at most four pure products. Since $\operatorname{rank} D = 4$, exactly four are necessary. We give a direct recovery formula using filtering normal forms [16, 17].

Bounded invertible filters R, S can be chosen so that

$$D' = (R\otimes S)D(R\otimes S)^*, \quad \frac{D'}{\tau} = \frac{1}{4}\left(I + \sum_{i=1}^{3} t_i\sigma_i\otimes\sigma_i\right), \quad \tau = \operatorname{Tr} D' > 0. \tag{7.5}$$

Their norms and inverse norms may all be bounded by four (Appendix D). Positivity of D' and $D'^\Gamma = (R\otimes\overline{S})K(R\otimes\overline{S})^*$ gives $\sum_i |t_i| \leq 1$.

Choose real unit vectors a, b with $a_i b_i = t_i$. For at least two nonzero coordinates, relabel so $p = |t_1| > 0$, $q = |t_2| + |t_3| > 0$ and set

$$r = \frac{2p^2}{1 + p^2 - q^2 + \sqrt{[1-(p+q)^2][1-(p-q)^2]}}, \quad a_1 = \sqrt{r}, \quad a_i = \sqrt{(1-r)|t_i|/q}. \tag{7.6}$$

Put $b_i = t_i/a_i$ on nonzero coordinates and $b_i = 0$ otherwise. The remaining branches are stated in the appendix. Let $P(v) = (I + v\cdot\sigma)/2$ and $E_4 = \{+++, +--, -+-, --+\}$. Sign averaging gives

$$D'/\tau = \frac{1}{4}\sum_{\epsilon\in E_4} P(\epsilon \odot a) \otimes P(\epsilon \odot b).$$

For corresponding unit spinors u_ϵ, v_ϵ, set $f_\epsilon = R^{-1}u_\epsilon$, $g_\epsilon = S^{-1}v_\epsilon$. Then

$$D = \frac{\tau}{4}\sum_{\epsilon\in E_4} f_\epsilon f_\epsilon^* \otimes g_\epsilon g_\epsilon^*, \qquad q(UAV^*) = \frac{\tau}{4}\sum_{\epsilon\in E_4} |f_\epsilon^* A g_\epsilon|^2. \tag{7.7}$$

Partial transpose replaces g_ϵ by $\overline{g_\epsilon}$ in the vector factor of K; row-first vectorization gives precisely $f_\epsilon^* A g_\epsilon$ in the scalar. The four terms are nonnegative and may vanish simultaneously. Compactness establishes filter existence; these formulas recover the products once the filters are chosen. A certified general-purpose filter solver is not supplied.

8 Exact controls in the complex nonnormal regime

The following family tests independent shores, complex phases and full local supports. Put $d_t = 1 + 2t^2$, $d_s = 1 + 2s^2$ for $t, s > 0$, and define orthonormal columns

$$\begin{aligned} x_t &= (|000\rangle + t|111\rangle + it|222\rangle)/\sqrt{d_t}, & w_t &= (|100\rangle + t|212\rangle + it|021\rangle)/\sqrt{d_t}, \\ y_s &= (|000\rangle + is|111\rangle + s|222\rangle)/\sqrt{d_s}, & v_s &= (|100\rangle + is|212\rangle + s|021\rangle)/\sqrt{d_s}. \end{aligned} \tag{8.1}$$

Let $X_t = [x_t, w_t]$, $Y_s = [y_s, v_s]$ and $C_{t,s} = X_t Y_s^*/\sqrt{2}$. Exact partial traces give

$$q(C_{t,s}) = \frac{11s^2t^2 + 8s^2 + 8t^2}{4d_t d_s}, \quad X_t^* Y_s = \frac{I_2}{\sqrt{d_t d_s}}, \quad \|C_{t,s}C_{t,s}^* - C_{t,s}^* C_{t,s}\|_F^2 = 1 - \frac{1}{d_t d_s}. \tag{8.2}$$

Every local range and corange support is $\mathbb{C}^3$. Neither global support plane contains a product vector: flatten a nonzero $ax_t + bw_t$ across the first site. For $a, b \neq 0$, row 2 has private columns $22, 12$, and rows 0,1 have private columns $21, 11$, respectively. If either coefficient is zero, all three rows still have distinct nonzero columns. The flattening has rank three in every case. The same reasoning applies to Y_s.

At $t = s = 1$, the normalized invariants are $(1, 1, 10/9, 2/9)$, the energy is $3/4$, and the three single-site partial traces have ranks $(5, 4, 6)$. Replacing the adjoint Y^* incorrectly by Y^T changes the energy to $25/36$. This supplies a sensitive exact phase and rank diagnostic.

For $U = X_1$, $V = Y_1$, direct swap compression gives

$$D = \frac{1}{72}\begin{pmatrix} 53 & 0 & 0 & 0 \\ 0 & 40 & 1+4i & 0 \\ 0 & 1-4i & 40 & 0 \\ 0 & 0 & 0 & 53 \end{pmatrix}. \tag{8.3}$$

Its spectrum is $\{53/72, 53/72, (40 - \sqrt{17})/72, (40 + \sqrt{17})/72\}$. Partial transpose moves the off-diagonal entries to positions $00, 11$ and $11, 00$, yielding spectrum $\{5/9, 5/9, (53 - \sqrt{17})/72, (53 + \sqrt{17})/72\}$. The unitary phase filter

$$R = \mathrm{diag}((1-4i)/\sqrt{17}, 1), \qquad S = I$$

gives $\tau = 31/12$ and $t = (\sqrt{17}/93, \sqrt{17}/93, 13/93)$ in (7.5). An irrelevant scalar phase can enforce determinant one. For $A = I_2/\sqrt{2}$, both (7.2) and (7.7) give 3/4.

For comparison, take $U = V = (|000\rangle, |100\rangle)$. Then

$$D = \tfrac{1}{4}(I - F_{\log}/2), \qquad K = \tfrac{1}{4}(I - |I_2\rangle\langle I_2|/2).$$

Here $\operatorname{rank} D = 4$ while $\operatorname{rank} K = 3$. The four tetrahedral Bloch vectors $n_\epsilon = \epsilon/\sqrt{3}$ give

$$D = \frac{3}{16} \sum_{\epsilon \in E_4} P(n_\epsilon) \otimes P(-n_\epsilon).$$

Antipodal spinors are orthogonal, so $A = I_2/\sqrt{2}$ annihilates all four terms. Direct calculation for arbitrary complex A gives

$$q(UAV^*) = \frac{|A_{00} - A_{11}|^2}{8} + \frac{|A_{01}|^2 + |A_{10}|^2}{4}.$$

Thus minimal product length four does not mean that four is the minimal number of arbitrary linear squares for K.

These controls support definitions and implementations. Their positivity is not evidence replacing the full Gram identity.

9 Discussion

The endpoint proof combines an intrinsic polynomial representation with an exact finite construction of its positive coefficients. The compound transport is short and preserves both physical shores, whereas the positive Gram construction contains the substantial finite computation. The distinction identifies two routes for further simplification: a more compact formula for the existing degree-five operators, or a new lower-degree identity.

A cubic representation of Δq has a precise two-sector form, but the holomorphic sector alone is obstructed by an exact negative direction in its linear carrier [20]. Appendix F records the statement and explains why that direction is not a physical feature. No mixed cubic construction is used in this paper.

The sharp qutrit threshold is a finite-copy result. It does not establish an NPT state undistillable at every copy number. Nor does the dimension-independent Cauchy–Binet transport supply positive operators in dimensions for which none have been constructed. Classification of all zero orbits, unrestricted higher-dimensional endpoints and all-copy questions are separate mathematical tasks.

Data availability and verification status

The computational proof objects currently reside in the local certificate collection identified in Appendix E. The recorded manifest lists 1779 data files, totaling 632802694 bytes, and 79 repository dependencies. A read-only inventory for this draft found every listed file and matching SHA256 hashes. No public archive identifier has been assigned in this draft.

The source record reports full physical transfer enclosures and earlier exact replays. Its joined receipt reuses saved large products; this draft does not relabel that receipt as a fresh replay. The present work independently checked the nine selected physical zeros and both full ambient stationarity products, the complex nonnormal family and paired matrix, and the improved scalar comparisons using exact arithmetic. It did not replay the large face/range, trial/defect or spectral certificates. The analytic implications are complete conditional on the finite predicates in Proposition 6.

Author and preparation statement

The provisional author byline is Physical Superintelligence. AI-assisted tools were used in preparing this draft for literature retrieval, organization, exposition, exact-check code and typesetting. This statement describes draft preparation and does not certify the contribution history of the underlying proof record. No independent referee review or formal proof-assistant verification is claimed.

A Carrier projectors and full transpose coordinates

For a partition λ of r, the central projector is

$$E_\lambda = \frac{d_\lambda}{r!} \sum_{\pi \in S_r} \chi_\lambda(\pi) U_\pi.$$

For $\lambda = (3,2)$ in S_5 the prefactor is $1/24$; for $(2,2)$ in S_4 it is $1/12$. In both cases

$$\chi_\lambda(\pi) = \frac{c_1(c_1-3)}{2} + c_2,$$

where c_1, c_2 count fixed points and two-cycles. Indeed the character is the permutation character on two-element subsets minus the permutation character on singletons. These formulas make (4.1) finite permutation expressions.

Here is a direct carrier-membership proof. The tensor $(e_0 \wedge e_1) \otimes (e_0 \wedge e_1)$ is a highest-weight vector of weight $(2,2)$: every raising matrix unit annihilates it. Its $GL(\mathcal{H})$ orbit spans the Cartan component $S_{(2,2)}\mathcal{H}$. Every decomposable $p \otimes p$ is in that component. Similarly,

$$e_0 \otimes (e_0 \wedge e_1) \otimes (e_0 \wedge e_1)$$

is a highest-weight vector of weight $(3,2)$. The same expression with e_1 in the first slot is obtained within its generated module by the lowering action. Every $c \otimes p \otimes p$, $c \in \mathrm{span}\{x_0, x_1\}$, is a transformed linear combination of these tensors. This proves the h inclusion, including degenerate $p = 0$. The mixed inclusion follows directly because the wedge map annihilates $c \otimes p$, whose orthogonal projector in $\mathcal{H} \otimes \bigwedge^2 \mathcal{H}$ is $A_{12} - \mathrm{Alt}_{012}$. Conjugating the $(2,2)$ argument proves the a inclusion. Only membership is required; arbitrary vectors in a carrier need not be features of one matrix.

For an explicit transpose convention, index the six-slot operator by (r, i, j, k, l, c) and (r', i', j', k', l', c'). The map Γ_{34} interchanges (k, l) with (k', l') between the operator's row and column indices. It leaves r, c, r', c' untouched. In particular it is the transpose of the whole exterior factor, realized first in the full tensor basis. Compressing first and transposing arbitrary native coordinates would not define the same map.

Let a block's uncompressed native embedding be $V = U_\tau JK$, where U_τ is a tensor product of local frames, J a site stabilizer image and K a forced-kernel frame. Its metric is $H_K = K^* J^* G_\tau JK$, $G_\tau = U_\tau^* U_\tau$. For a positive definite dyadic preconditioner metric, an exact orthonormal image frame is

$$F = JKW \ \mathrm{chol}(W^* H_K W)^{-*}, \qquad F^* G_\tau F = I,$$

where $\mathrm{chol}(H)\,\mathrm{chol}(H)^* = H$. For a complement, first form $W_\perp = W_0 - K H_K^{-1} K^* J^* G_\tau J W_0$ over the rationals and then whiten it in its own metric. These formulas separate the exact frame from a numerical proposal for the dyadic preconditioner.

An off-diagonal native atom gives a useful normalization check. For individually normalized real native vectors v_i, v_j with $\rho = \langle v_i, v_j \rangle$, the symmetric atom $v_i v_j^* + v_j v_i^*$ has HS norm $\sqrt{2 + 2\rho^2}$.

Dividing by $\sqrt{2}$ would be incorrect unless the vectors are orthogonal. Physical multiplicity contributes a further $\sqrt{\nu}$ in orthonormal operator coordinates.

Writing the local metric as $G_{s,t} = L_{s,t}L_{s,t}^*$, the full signed reverse map R_s^{-1} has normalized local blocks

$$R_s^{\mathrm{norm}}[a,b] = (L_{h,a}^{-1} \otimes L_{h,a}^{-1}) R_s^{-1}[a,b](L_{s,b} \otimes L_{s,b}) \sqrt{\frac{d_{h,a}}{d_{s,b}}}. \tag{A.1}$$

Here $d_{s,b}$ is the local irreducible representation dimension. The complete global contraction uses two target frames, all three local maps and two source frames:

$$abcx,\ defy,\ adgj,\ behk,\ cfil,\ ghiu,\ jklv \longmapsto xyuv.$$

Sum the six site permutations with the source stabilizer divisor; retain both standard representation copies, including $(\sigma - \sigma^2)Y/\sqrt{3}$. The remaining weight is

$$\frac{1}{\mathrm{stabilizer}_s} \sqrt{\frac{\deg_h \mathrm{orbit}_h}{\deg_s \mathrm{orbit}_s}},$$

together with the off-diagonal target-row factor and source-atom norm. These definitions identify the map whose physical singular values occur in Proposition 6; the complete block dictionary is finite data, not an additional invariance assumption.

B Explicit physical zeros used for range relations

Let $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $J_r = \epsilon^{\otimes r}$. Embed binary words lexicographically into the corresponding qutrit coordinates. The selected binary points have the forms

$$\begin{aligned} \text{I:} \quad & X = I_2 \otimes \chi, \quad B = I_2 \otimes \beta^T, \\ \text{II:} \quad & X = a \otimes [u,v], \quad B = \beta^T \otimes ([v^T; -u^T] J_2^{-1}), \\ \text{III:} \quad & X = [u,v], \quad B = [v^T; -u^T] J_3^{-1}. \end{aligned} \tag{B.1}$$

The transpose in the spin formula is part of its definition; the actual corange remains $\operatorname{ran} B^*$. The following literal Gaussian-integer vectors define the seven selected binary points. Parentheses denote columns.

III-1

$$\begin{aligned} u &= (1,0,1,i,1+i,0,-1,1)^T \\ v &= (0,1,i,1,0,1-i,1,-1)^T \end{aligned}$$

III-2

$$\begin{aligned} u &= (-1+i,-1,-1+i,i,i,i,-1,-1-i)^T \\ v &= (-1+i,1+i,-1+i,-1-i,i,1-i,i,-i)^T \end{aligned}$$

III-3

$$\begin{aligned} u &= (-1-i,-1+i,-1-i,-1+i,-1-i,-1,-1-i,-1)^T \\ v &= (-i,-i,-i,-i,i,1+i,-1-i,i)^T \end{aligned}$$

III-4

$$u = (i, 1-i, -1-i, 0, -i, -1, 1-i, i)^T$$
$$v = (1, -1-i, -1-i, -1-i, -1, 0, 1+i, -1)^T$$

I-1

$$\chi = (i, -1-i, -1+i, -1+i)^T$$
$$\beta = (-1-i, -1-i, 1, -1-i)^T$$

II-1

$$a = (-1, -1+i)^T$$
$$\beta = (0, -i)^T$$
$$u = (-1-i, -1, 1-i, 1+i)^T$$
$$v = (-i, 1-i, 0, 1-i)^T$$

I-2

$$\chi = (1, 0, 1-i, 0)^T$$
$$\beta = (1, -i, 1-i, -1)^T$$

For the two qutrit spectator points use $X = U_2 \otimes \chi$, $B = U_2^T \otimes \langle 00|$, where $U_2 = (e_0, e_1)$ and χ is the vectorization of one of

$$\begin{pmatrix} i & 0 & -1+i \\ 0 & 1-i & 0 \\ 0 & 1+i & i \end{pmatrix}, \qquad \begin{pmatrix} -1-i & 1 & 0 \\ 1-i & -i & -1-i \\ 1+i & 1-i & 0 \end{pmatrix}.$$

For every one of the nine matrices $C = XB$, exact arithmetic verifies

$$\operatorname{rank} X = 2, \qquad q(C) = 0, \qquad X^* \mathcal{L}(C) = 0, \qquad \mathcal{L}(C) B^* = 0 \tag{B.2}$$

in all 27 physical coordinates. The accompanying small-check script reconstructs these matrices from the printed specifications.

For derivatives, write $w = x_0 \otimes x_1 - x_1 \otimes x_0 = \sqrt{2}p$. Then $\phi_h = Cww = 2z_h$ and

$$\dot{C} = \dot{X}B + X\dot{B}, \quad \dot{w} = \dot{x}_0 \otimes x_1 + x_0 \otimes \dot{x}_1 - \dot{x}_1 \otimes x_0 - x_1 \otimes \dot{x}_0,$$

$$\dot{\phi}_h = \dot{C}ww + C\dot{w}w + Cw\dot{w}.$$

The full derivative $\dot{\phi}_h \phi_h^* + \phi_h \dot{\phi}_h^*$ must be formed before symmetry averaging. Both real and imaginary factor directions contribute. Equation (B.2) makes its target pairing zero; the factor of four relative to normalized features is retained.

The face construction uses covariances of the physical zero families and their local-unitary, site-permutation and conjugation closures. The nine points supply selected derivative relations; they are not claimed by themselves to classify every zero or determine every forced-face dimension. Verifying the full covariance kernels and the rank of their coefficient relations remains part of the finite certificate, distinct from (B.2).

C A direct proof of the partial-trace bound

Set $\Phi(D) = (\mathrm{Tr}_1 D, \mathrm{Tr}_2 D)$ for a bipartite operator. Swap expansion gives

$$\|\Phi(uv^*)\|^2 \le \|u\|^2\|v\|^2 + |v^*u|^2,$$

because the difference is the expectation of $(I-F_1)(I-F_2) \succeq 0$ on $u\otimes v$. Rectangular reshaping and cyclicity of trace give

$$\langle \Phi(uv^*), \Phi(st^*)\rangle = \langle \Phi(tv^*), \Phi(su^*)\rangle.$$

In detail, if the reshaped matrices are U, V, S, T, the two partial traces of uv^* are $(V^*U)^T$ and UV^*; the two scalar contractions exchange their roles.

Use an SVD factorization $D = XY^*$ with $X^*X = Y^*Y = \Sigma$, padded to two columns. Write $Y^*X = \tau I + (a+ib)\cdot\sigma$, $\tau = \mathrm{Tr}\, D/2$, $a, b \in \mathbb{R}^3$. A unit Bloch axis orthogonal to both a, b gives a unitary basis with equal diagonal entries τ. Rotate both factors by that unitary. Their columns satisfy

$$x_i^* x_j = y_i^* y_j = g_{ij}, \qquad y_i^* x_i = \tau.$$

Writing $\delta_i = g_{ii}$ and $g = g_{12}$, the dyad estimate, the exchange identity and Cauchy–Schwarz imply

$$\|\Phi(D)\|^2 \le (\delta_1+\delta_2)^2 + 2|g|^2 + 2|\tau|^2.$$

On the other hand, $\|D\|_F^2 = \delta_1^2 + \delta_2^2 + 2|g|^2$ and $|\mathrm{Tr}\, D|^2 = 4|\tau|^2$. Subtracting leaves $(\delta_1 - \delta_2)^2 + 2|g|^2 \ge 0$, which proves (6.2). Rank loss is covered by padding; the scalar ambient case is immediate.

D Filtering and every product-recovery branch

The bound $\mathrm{Tr}\, D \le 4$ follows because each diagonal element of D is a unit-dyad energy, and $\mathcal{L} \preceq I$: the eigenvalues of each local map are $1, -1/2$. Minimize $\mathrm{Tr}[(R\otimes S)D(R\otimes S)^*]$ over $\det R = \det S = 1$. For $X = R^*R$, $Y = S^*S$, the objective is at least $\mathrm{Tr}\, X\, \mathrm{Tr}\, Y/8$, and the identity filters give at most four. Since $\det X = \det Y = 1$, one has

$$\mathrm{Tr}\, X, \mathrm{Tr}\, Y \ge 2, \qquad \mathrm{Tr}\, X\, \mathrm{Tr}\, Y \le 32, \qquad \mathrm{Tr}\, X, \mathrm{Tr}\, Y \le 16.$$

Thus all filter singular values and inverse singular values are at most four on the sublevel. Compactness gives a minimizer. Traceless Hermitian left variations force scalar marginals. Proper rotations, implemented by logical unitaries, diagonalize the real Pauli correlation matrix with signed singular values, yielding (7.5). The positive background transforms to $(RR^*\otimes SS^*)/8$; it is not reset to $I/8$.

For $p, q > 0$, $\kappa = p + q \le 1$, the root in (7.6) also equals

$$\frac{1 + p^2 - q^2 - \sqrt{[1-(p+q)^2][1-(p-q)^2]}}{2}.$$

The rationalized expression avoids subtraction of nearly equal positive terms. It has $0 < r < 1$ and solves

$$p^2/r + q^2/(1-r) = 1.$$

Thus the resulting a, b are unit vectors with $a_i b_i = t_i$. At $\kappa = 1$, $r = p$ and no nonzero coordinate has a singular denominator. If only t_i is nonzero, choose $a = e_i$, $b = t_i e_i + \sqrt{1-t_i^2}\, e_j$, $j \ne i$; if all vanish, choose $a = e_1, b = e_2$. Some octahedral edges and vertices correspond to singular

D' and cannot arise from invertible filtering of the physical $D > 0$, but remain valid algebraic branches.

For a unit Bloch vector (x, y, z), a corresponding spinor is

$$\left(\sqrt{\frac{1+z}{2}},\ \frac{x+iy}{\sqrt{2(1+z)}}\right)^T \quad (z \neq -1),$$

and $(0,1)^T$ for $z = -1$. The sign average over E_4 has zero single signs and $\frac{1}{4}\sum_\epsilon \epsilon_i\epsilon_j = \delta_{ij}$. Expanding both Pauli factors therefore proves (7.7), including its complex phase and boundary conventions.

E Finite certificate predicates and provenance

The following table separates the mathematical predicates from the availability of a saved receipt. Each stage uses exact data or rigorous enclosures of the defined physical objects. A complete checker must establish all predicates needed for Proposition 6.

Stage	Required mathematical predicate
Geometry	Carrier and face projectors are idempotent and physically self-adjoint; their inclusions, covariance kernels, dimensions and block multiplicities agree with the complete tensor definitions.
Range	$L\mathsf{A} = 0$ and $Lt = 0$ over $\mathbb{Q}$. The designated 181- and 2282-order minors are nonzero modulo 65521 with invertible denominators. The independent transfer columns belong to $\mathcal{U}$.
Trial	Each native factor Z_b has the prescribed full rank on its face; $V_b Z_b Z_b^T V_b^*/2^{160}$ exceeds its weighted positive margin in the original frame metric.
Defect	Reconstruct both complete transpose outputs and all 122 target blocks. The physical HS norm obeys $\mathrm{Tr}(E^2) < 2^{-70}$. Residue reconstruction includes a uniqueness bound.
Transfer	Source and output dictionaries are physical isometries. Every one of the $8 \cdot 83 = 664$ block enclosures covers its full contraction, and their aggregate gives $\|\mathcal{D} - A_{48}\|_F^2 < 2^{-64}$.
Spectrum	Exact products establish the grid rounding, approximate orthogonality and full residual of the specified dyadic spectral witness. The rational polar bound proves $\lambda_{182}(S_{48}) > 2^{-30}$.
Closure	Only the preceding established outputs enter Lemmas 7 and 5. The rational correction satisfies every complete target equation and the corrected face margins in Table 1.

The correction domain has three full symmetric blocks of sizes $97, 98, 55$ and five singleton atoms: $4753 + 4851 + 1540 + 5 = 11149$. In the source dictionary these are blocks $m346$-0, $m456$-0, $a146$-0, and the selected atoms in $m006$-0, $m036$-0, $m116$-0, $m444$-0, $m555$-0. The designated independent columns occur with counts $22, 1687, 568, 1, 1, 1, 1, 1$. The 2463 off-face outputs include every mixed face/complement entry and every complementary entry; supported h coordinates are eliminated and later restored, not discarded.

For the spectral witness, let U, Λ be the specified square dyadic matrix and increasing dyadic diagonal. Establish

$$\|U^*U - I\|_{\mathrm{op}} \leq \eta < 1/2, \qquad \|S_{48}U - U\Lambda\|_{\mathrm{op}} \leq \xi.$$

For the exact polar unitary $Q = U(U^*U)^{-1/2}$, $\|Q - U\|_{\mathrm{op}} \le \eta$, whence

$$\|S_{48} - Q\Lambda Q^*\|_{\mathrm{op}} \le \xi + (\|S_{48}\|_F + \|\Lambda\|_{\mathrm{op}})\eta. \tag{E.1}$$

The saved spectral receipt gives rational upper bounds $\eta < 2^{-35}$, $\xi < 2^{-35}$ and a total bound in (E.1) below 2^{-33}; its actual rational values are retained in the accompanying metadata. The required comparison with the 182nd diagonal entry is part of the spectral predicate. A recorded bound is not substituted for checking the products that establish it.

For denominator-cleared integer reconstruction, a bound $|n| \le B$ and modulus product $P > 2B$ make the symmetric CRT representative unique. Rational reconstruction instead requires its own numerator, denominator and coprimality conditions. The residue values must be tied to the defined full contractions. Reproducing saved residues alone establishes a smaller claim.

The interval implementation uses directed rounding, finite centers and nonnegative radii, with fast-math and fused contraction disabled. The arithmetic controls must exercise hardware rounding and subnormal behavior. In the source's integer-limb spectral products, each integer dot product is justified by inner dimension $\cdot \max|a| \cdot \max|b| \le 2^{53}$; signed reassembly has a separate bound below 2^{126}. These bounds make the arithmetic exact under the documented implementation assumptions. They are not numerical convergence tests. Arb supplies midpoint-radius enclosures for algebraic frame normalization [18].

E.1 Source identification and checks performed for this draft

The source endpoint record is ZD359/N3-379, its intrinsic reformulation ZD360/N3-380. The original certificate manifest has SHA256

```
1407698fb15cd9a630f95ca18d6778a3
dc31f8e6a4baa34c8c0020a65af2cbbf.
```

The recorded package has 1779 files and 79 additional repository dependencies. Every listed file was present and matched this manifest during preparation of the draft. This check did not evaluate the mathematical predicates in the table.

The supplementary draft metadata contain the inventory, the original spectral receipt, the nine literal point specifications and an independent small exact-check script. The script checks all nine instances of (B.2), the family energy (8.2), its partial-trace ranks, the complex paired operator (8.3), the phase normal form and the rational improved-gap comparisons. It does not verify the large covariance, range, defect, interval-transfer or spectral objects.

The original joined receipt explicitly reuses saved large products, while the source narrative reports an earlier fresh replay. Neither statement is presented as a new external replay. Before journal submission, the complete closed proof collection requires a stable accessible release with its code/data dictionary, arithmetic environment, exact commands and a clear account of which expensive predicates are recomputed and which are established from retained witnesses. No such release URL is invented here.

F The boundary of cubic simplification

The degree-five identity (3.4) is sufficient for the endpoint. A lower-degree candidate for Δq has features $w_h = \operatorname{vec} C \otimes \operatorname{vec} M(C)$ and $w_m = \operatorname{vec} C \otimes \overline{\operatorname{vec} M(C)}$. Order their slots as matrix row, minor row pair, minor column pair, matrix column. Set

$$A = A_{12}A_{34}, \qquad \Pi = (A_{12} - \mathrm{Alt}_{012})(A_{34} - \mathrm{Alt}_{345}).$$

Actual features satisfy $\Pi w_h = w_h$, $A w_m = w_m$. With Γ transposing the whole minor subsystem, the complete mixed cubic target is

$$G = AGA \succeq 0, \qquad \Pi[T_0 - \Gamma(G)]\Pi \succeq 0. \tag{F.1}$$

Phase averaging separates the four cubic bidegrees. Vanishing at rank one makes each holomorphic quadratic factor a combination of two-by-two minors; conjugation identifies the two pairs of bidegrees. The holomorphic coordinate functions span the incidence carrier, and their products with independently complexified conjugates are linearly independent after quotienting linear relations. Thus no extra Hermitian relation weakens (F.1).

The intrinsic source [20] exhibits a negative carrier vector using local pairing tensors b_σ, $\sigma \in S_3$, with metric $\langle b_\sigma, b_\tau \rangle = 3^{\#\mathrm{cycles}(\sigma^{-1}\tau)}$. For $w = \Pi(b_{\mathrm{id}} \otimes b_{\mathrm{id}} \otimes b_{(2,0,1)})$,

$$\|w\|^2 = 3132, \qquad \langle w, T_0 w \rangle = -321.$$

This excludes holomorphic cubic squares alone. It does not exclude the mixed sector. The flattening of w across the matrix and minor slots has a 2×2 minor $\mathrm{diag}(1/2, -1/12)$ of determinant $-1/24$, whereas every actual $C \otimes M(C)$ has rank-one flattening. The negative vector is therefore not a physical witness.

Finally, direct permutation expansion gives $\Gamma(\Pi) \succeq (4/9)A$. Any solution of (F.1) would obey the single trace budget

$$\operatorname{Tr} G_h^{(3)} + \operatorname{Tr}[G\Gamma(\Pi)] = 24843000, \qquad \operatorname{Tr} G_h^{(3)} + \tfrac{4}{9} \operatorname{Tr} G \le 24843000.$$

This bounds a feasible set but does not establish feasibility. Scalar padding $G = t\Gamma(\Pi)$ fails: for $t > 0$ it cannot annihilate the nonzero mixed features at physical endpoint zeros. The cubic discussion identifies a simplification target; it is not a dependency of the endpoint proof.

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